



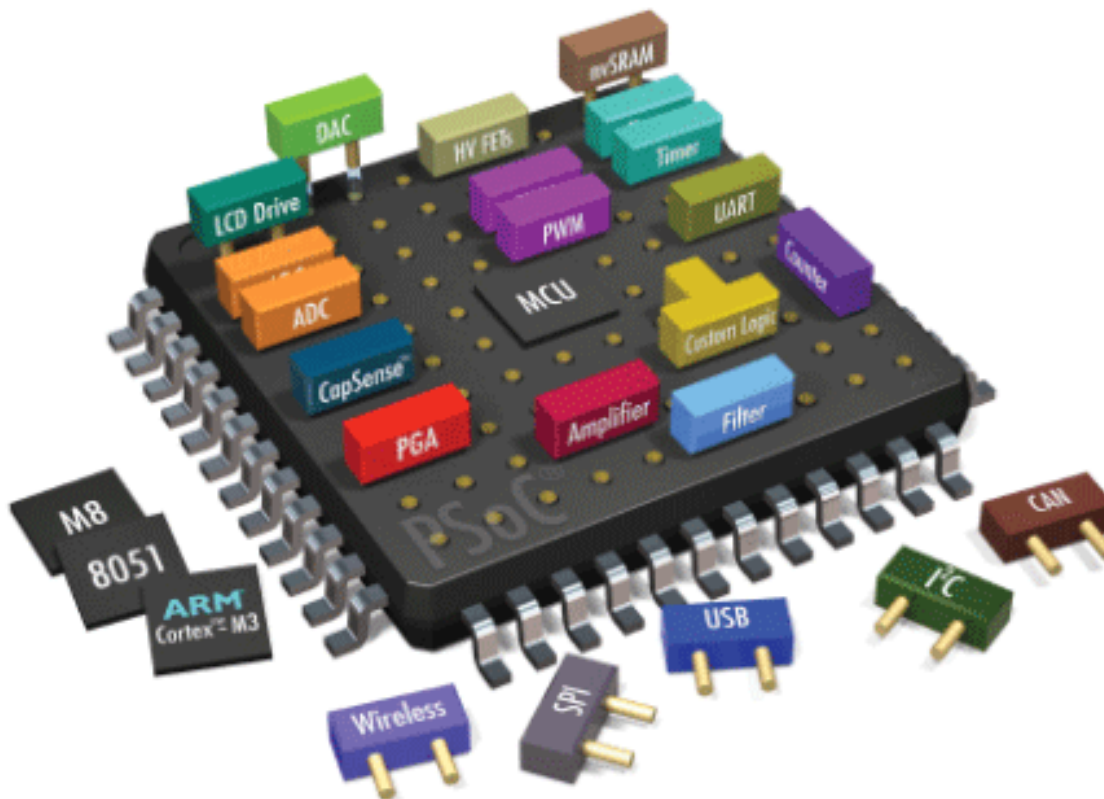
HITEC UNIVERSITY, TAXILA

DEPARTMENT OF COMPUTER ENGINEERING

Lab # 09: Demonstration and practice of delay generation, loop counter and use of lookup tables.

Embedded Systems

Demonstration and practice of delay generation, loop counter and use of lookup tables.



Dated:

Week 9



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Semester:

Fall 2023

8051 microcontroller:

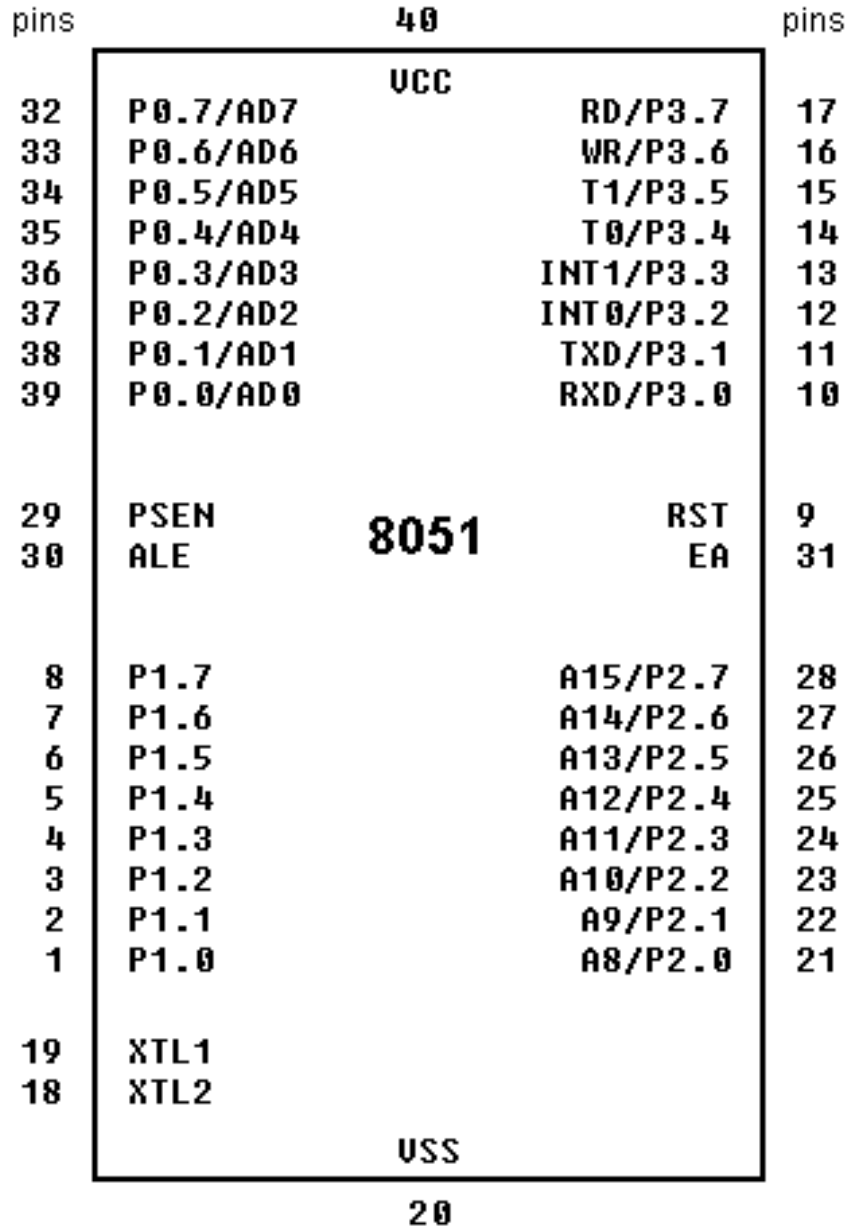
8051 microcontroller is designed by Intel in 1981. It is an 8-bit microcontroller. It is built with 40 pins DIP (dual inline package), 4kb of ROM storage and 128 bytes of RAM storage, 2 16-bit timers. It consists of are four parallel 8-bit ports, which are programmable as well as addressable as per the requirement. An on-chip crystal oscillator is integrated in the microcontroller having crystal frequency of 12 MHz.



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- **Pins 1 to 8** – These pins are known as Port 1. This port doesn't serve any other functions. It is internally pulled up, bi-directional I/O port.
- **Pin 9** – It is a RESET pin, which is used to reset the microcontroller to its initial values.
- **Pins 10 to 17** – These pins are known as Port 3. This port serves some functions like interrupts, timer input, control signals, serial communication signals RxD and TxD, etc.



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Pin	Name	Bit Address	Function
P3.0	RXD	B0H	Receive data for serial port
P3.1	TXD	B1H	Transmit data for serial port
P3.2	INT0-bar	B2H	External interrupt 0
P3.3	INT1-bar	B3H	External interrupt 1
P3.4	T0	B4H	Timer/counter 0 external input
P3.5	T1	B5H	Timer/counter 1 external input
P3.6	WR-bar	B6H	External data memory write strobe
P3.7	RD-bar	B7H	External data memory read strobe

- **Pins 18 & 19** – These pins are used for interfacing an external crystal to get the system clock.
- **Pin 20** – This pin provides the power supply to the circuit.
- **Pins 21 to 28** – These pins are known as Port 2. It serves as I/O port. Higher order address bus signals are also multiplexed using this port.
- **Pin 29** – This is PSEN pin which stands for Program Store Enable. It is used to read a signal from the external program memory.
- **Pin 30** – This is EA pin which stands for External Access input. It is used to enable/disable the external memory interfacing.
- **Pin 31** – This is ALE pin which stands for Address Latch Enable. It is used to demultiplex the address-data signal of port.
- **Pins 32 to 39** – These pins are known as Port 0. It serves as I/O port. Lower order address and data bus signals are multiplexed using this port.
- **Pin 40** – This pin is used to provide power supply to the circuit.

The Data Pointer:

The Data Pointer (DPTR) is the 8051's only user-accessible 16-bit (2-byte) register. The Accumulator, R0–R7 registers and B register are 1-byte value registers. DPTR is meant for pointing to data. It is used by the 8051 to access external memory using the address indicated by DPTR. DPTR is the only 16-bit register available and is often used to store 2-byte values.



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MOVC:

Move Code Byte to Accumulator. MOVC moves a byte from Code Memory into the Accumulator. The Code Memory address from which the byte will be moved is calculated by summing the value of the Accumulator with either DPTR or the Program Counter (PC). In the case of the Program Counter, PC is first incremented by 1 before being summed with the Accumulator.

Syntax: MOVC A,@A+register

Instructions
MOVC A,@A+DPTR
MOVC A,@A+PC

MOVC will store your data into the code memory.

MOV will store your data into the data memory.

MOVX will store your data into external memory (given that your microcontroller is interfaced with an external memory)

Example 1:

This Example uses the technique of table lookup to show the LED patterns resided in memory. Each pattern stays on 0.1 seconds with the help of delay generation technique.

Description:

The data transfer instruction “MOVC A,@A+DPTR” is used to read data bytes resident in the program memory. The data read is first loaded into accumulator A and then written to port 2. The loop counter R0 is loaded with 23 data bytes. Once a data byte is read and written, the contents of A are reset and the DPTR is incremented by one to point the next data byte until the loop counter is decreased to zero.

Code:

```
ORG 000H
```

```
START:
```

```
MOV R0,#23
```

```
MOV DPTR,#TABLE
```

```
NEXT:
```

Embedded Systems

Session: 2K20 Computer (7th Semester)

Lab Instructor: Bushra Fiaz



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```
MOV A,#0
MOVC     A,@A+DPTR
MOV P2,A
CALL DELAY
INC DPTR
DJNZ R0,NEXT
```

```
JMP START
```

```
=====
;
; DELAY 0.1S
;
=====
```

DELAY:

```
MOV R6,#200
```

DL1:

```
MOV R7,#249
DJNZ R7,$
DJNZ R6,DL1
RET
```

```
=====
;
TABLE: DB 00011000B
        DB 00100100B
        DB 01000010B
        DB 10000001B
        DB 01000010B
        DB 00100100B
        DB 00011000B
        DB 00H,FFH,00H,FFH
        DB 00H,FFH,00H,FFH
        DB 0FH,F0H,0FH,F0H
        DB 0FH,F0H,0FH,F0H
```

```
END
```

Example 2:

Turn on LEDs according to the following cases by using subroutines and repeat each case 8 times using loop counters.

- Case 1: Turn on LEDs from left to right.
- Case 2: Turn on LEDs from right to left.
- Case 3: Blink two sets (left and right sides) of LEDs alternately.



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- **Case 4: Blink all of 8 LEDs.**

Description:

The cases 1, 2, 3, and 4 are performed by the subroutines RIGHT, LEFT, SHOW1, and SHOW2, respectively. The R0 is a loop counter used to determine the number of subroutines call.

Code:

```
ORG 000H

START:
    MOV R0,#8
NEXT1:
    CALL RIGHT
    DJNZ R0,NEXT1

    MOV R0,#8
NEXT2:
    CALL LEFT
    DJNZ R0,NEXT2

    MOV R0,#8
NEXT3:
    CALL SHOW1
    DJNZ R0,NEXT3

    MOV R0,#8
NEXT4:
    CALL SHOW2
    DJNZ R0,NEXT4

    JMP START

=====
RIGHT:
    MOV A,#10000000B
N_R:
    MOV P2,A
    CALL DELAY
    RR A
    CJNE A,#10000000B,N_R
    RET
```



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```
;=====
```

```
;LEFT:
```

```
MOV A,#00000001B
```

```
N_L:
```

```
MOV P2,A
```

```
CALL DELAY
```

```
RL A
```

```
CJNE A,#00000001B,N_L
```

```
RET
```

```
;=====
```

```
;SHOW1:
```

```
MOV A,#00001111B
```

```
MOV P2,A
```

```
CALL DELAY
```

```
SWAP A
```

```
MOV P2,A
```

```
CALL DELAY
```

```
RET
```

```
;=====
```

```
;SHOW2:
```

```
MOV A,#FFH
```

```
MOV P2,A
```

```
CALL DELAY
```

```
CPL A
```

```
MOV P2,A
```

```
CALL DELAY
```

```
RET
```

```
;=====
```

```
;DELAY 0.1S
```

```
;=====
```

```
;DELAY:
```

```
MOV R6,#200
```

```
DL1:
```

```
MOV R7,#249
```

```
DJNZ R7,$
```

```
DJNZ R6,DL1
```

```
RET
```




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END

Example 3:

Display the alphabet A on Dot Matrix LED (5*7).

Description:

The pattern data (five columns) of the alphabet “A” is stored in TABLE are read and written to P2. While the pattern data of a column is emitted from P2. The scan data (a 1) is sent to P0 to turn on the LEDs in the corresponding column, The operation procedure of a scanned LED display is also used in the dot matrix KD scanning. The steps are:

1. Send pattern data of a column to the column lines.
2. Scan the column whose LEDs to be turned on.
3. Delay an appropriate time.
4. Turn off the LEDs in the column if necessary.

By repeating this procedure for each pattern data, the character of “A” is thus displayed on the dot matrix LED.

Code:

```
ORG 000H
```

```
CLR P1.7
```

START:

```
MOV DPTR,#TABLE
```

```
MOV R2,#10000000B
```

```
MOV R1,#0
```

NEXT:

```
MOV A,R1
```

```
MOVC A,@A+DPTR
```

```
MOV P2,A
```

```
MOV A,R2
```

```
MOV P0,A
```

```
CALL DELAY
```

```
RR A
```

```
MOV R2,A
```

```
INC R1
```



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```
CJNE R1,#5,NEXT

JMP START

;=====
; DELAY 5mS
;=====
DELAY:
    MOV R6,#10
DL1:
    MOV R7,#249
    DJNZ R7,$
    DJNZ R6,DL1
    RET

;=====
TABLE:    DB    3EH,48H,88H,48H,3EH

END
```

Lab Tasks:

- 1. Rewrite task 2 of lab 8 using table lookup. The task is: Write a program which act as up counter from 00 to 99 using the two 7-segment displays (DS3 and DS4). Each number stays on 0.1 seconds.**
 - 2. Rewrite task 3 of lab 7 using table lookup. The task is: Write a program which display the first alphabet of the name of all group members on Dot Matrix LED. Each alphabet stays on 0.5 seconds.**
Hint: For this purpose, use Delay given in example 3. The time of 25ms is needed to display a character (5 columns) since the 5-ms delay subroutine is called for one column display. This speed is too fast to display character visibly. A way to display a visible character is that the character is displayed several times. For this purpose, use loop counter technique. For example, the register R3 can be initiated to 20 to display character 20 times. Therefore, a character stays $25\text{ms} \times 20 = 0.5\text{s}$.
- Task 2 is done as your lab report.

Lab Performance Evaluation:

Read and practice the manual. Performance is evaluated at the end of lab based on successfully running and understanding of examples, tasks and viva using defined rubrics.



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Lab Report Evaluation:

Submit the Lab Report by writing all the examples and tasks which must include the following:

1. Brief description about how its work (3-5 lines)
2. The code
3. The results (Screenshot)

Important note: Lab Report must be according to the Lab Report Format available on Google Classroom and must be submitted on time. Two similar reports will get zero marks. So, don't try to copy your fellow reports. Lab report must be submitted in group of three maximum.